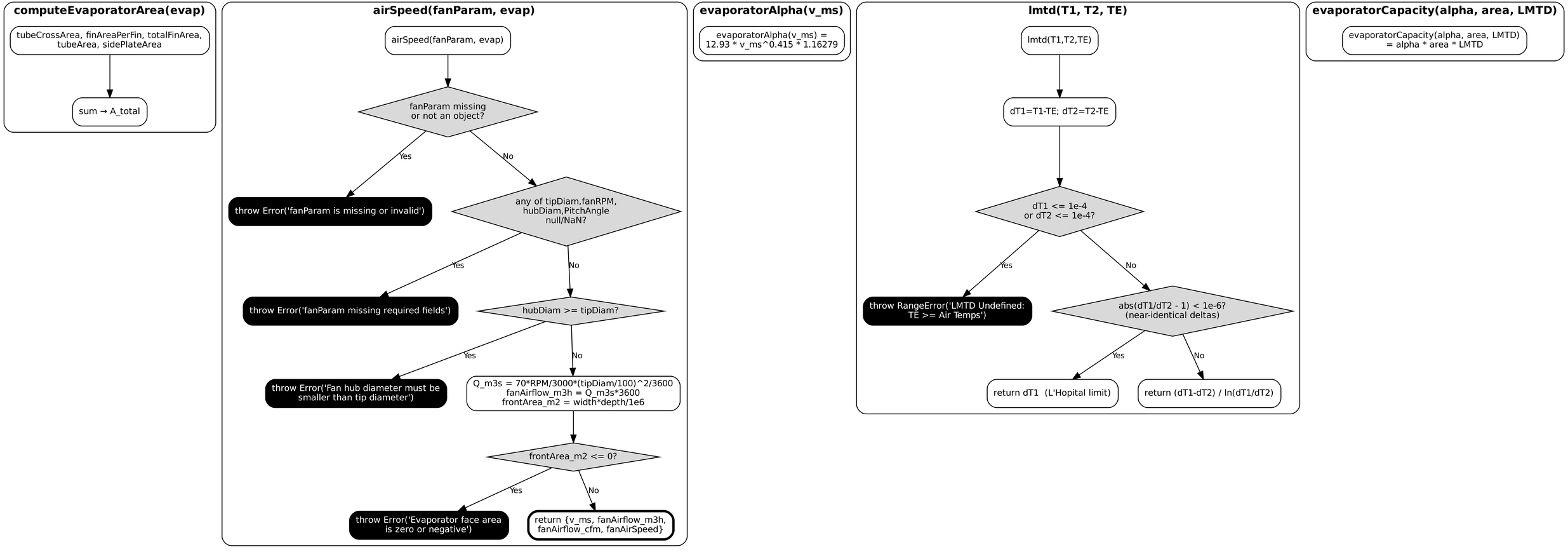
computeEvaporatorArea(evap)
tubeCrossArea, finAreaPerFin, totalFinArea, tubeArea, sidePlateArea
sum → A_total
airSpeed(fanParam, evap)
airSpeed(fanParam, evap)
fanParam missing or not an object?
Yes
No
throw Error('fanParam is missing or invalid')
any of tipDiam,fanRPM, hubDiam,PitchAngle null/NaN?
Yes
No
throw Error('fanParam missing required fields')
hubDiam >= tipDiam?
Yes
No
throw Error('Fan hub diameter must be smaller than tip diameter')
Q_m3s = 70*RPM/3000*(tipDiam/100)^2/3600
fanAirflow_m3h = Q_m3s*3600
frontArea_m2 = width*depth/1e6
frontArea_m2 <= 0?
Yes
No
throw Error('Evaporator face area is zero or negative')
return {v_ms, fanAirflow_m3h, fanAirflow_cfm, fanAirSpeed}
evaporatorAlpha(v_ms)
evaporatorAlpha(v_ms) = 12.93 * v_ms^0.415 + 1.16279
lmtd(T1, T2, TE)
lmtd(T1,T2,TE)
dT1=T1-TE; dT2=T2-TE
dT1 <= 1e-4 or dT2 <= 1e-4?
Yes
No
throw RangeError('LMTD Undefined: TE >= Air Temps')
abs(dT1/dT2 - 1) < 1e-6? (near-identical deltas)
Yes
No
return dT1 (L'Hopital limit)
return (dT1-dT2) / ln(dT1/dT2)
evaporatorCapacity(alpha, area, LMTD)
evaporatorCapacity(alpha, area, LMTD) = alpha * area * LMTD